Tackling data scarcity in machine learning-based CFRP drilling performance prediction through a broad learning system with virtual sample generation (BLS-VSG)

Hot Paper

By Ge, J (Ge, Jia) [1] , [2] ; Yao, ZQ (Yao, Zequan) [3] ; Wu, M (Wu, Ming) [3] , [4] ; Almeida, JHS Jr (Almeida Jr, Jose Humberto S.) [5] ; Jin, Y (Jin, Yan) [1] ; Sun, D (Sun, Dan) [1]

View Web of Science ResearcherID and ORCID (provided by Clarivate)

Source COMPOSITES PART B-ENGINEERING

View Journal Impact

Volume: 305

DOI: 10.1016/j.compositesb.2025.112701

Article Number 112701

Published OCT 2025

Early Access JUN 2025

Indexed 2025-07-03

Document Type Article

Jump to Enriched Cited References

Abstract Machine learning (ML)-based data-driven method has emerged as a powerful tool for predicting the manufacturing performance of carbon fibre reinforced plastic (CFRP), particularly in CFRP machining, where physics-based models are computationally expensive. However, the effectiveness of ML models are often constrained by limited datasets, due to the high cost and time required for experimental data acquisition. To address this, this paper presents the first study to apply virtual sample generation (VSG) techniques to enlarge the training dataset and mitigate data scarcity in the prediction of CFRP drilling performance. A novel hybrid ML framework integrating Broad Learning System (BLS) and VSG (BLS-VSG) is proposed to combine the capability of BLS in small dataset prediction with the enlarged dataset generated by VSG. The model has been employed to predict the drilling thrust force and delamination damage under various drilling conditions (spindle speed, feed rate, point angle). Three different VSG methods (SMOTE, MD-MTD and CVT) and the number of virtual samples were evaluated in detail. Results show that VSG can effectively enlarge the training dataset and improve the prediction performance of the ML model. Specifically, VSG reduced the mean square error (MSE) and mean absolute percentage error (MAPE) for thrust force prediction by 39.0 % and 12.9 %, respectively, compared to the benchmark without VSG. For delamination factor Fda prediction, MSE and MAPE were reduced by 22.6 % and 16.5 %, respectively. The proposed BLS-VSG model outperforms other conventional ML models (BPNN, ELM, SVR and RT) for both scenarios (with/without VSG), providing a robust and data-efficient solution for CFRP drilling

performance prediction.

Keywords

Author Keywords: Carbon Fibre Reinforced Plastic (CFRP); Drilling; Machine Learning (ML); Data-driven method; Virtual Sample Generation (VSG)

Keywords Plus: THRUST FORCE; DELAMINATION; OPTIMIZATION; NETWORK; SPEED

Author Information

Corresponding Address: Wu, Ming (corresponding author)

KULeuven, Dept Mech Engn, B-3001 Leuven, Belgium

E-mail Addresses :
ming.wu1@kuleuven.be

Addresses :

[1] Queens Univ Belfast, Sch Mech & Aerosp Engn, Belfast BT9 5AH, North Ireland

[2] Univ Coll Dublin, Sch Mech & Mat Engn, Dublin D04 V1W8, Ireland

[3] KULeuven, Dept Mech Engn, B-3001 Leuven, Belgium

[4] Katholieke Univ Leuven, Dept Comp Sci, B-3001 Leuven, Belgium

[5] LUT Univ, Dept Mech Engn, Lappeenranta, Finland

E-mail Addresses :
ming.wu1@kuleuven.be

Data availability statement

Data will be made available on request.

Categories/ Classification

Research Areas: Engineering; Materials Science

Citation Topics: 7 Engineering & Materials Science › 7.227 Manufacturing › 7.227.355 Advanced Machining

Sustainable Development Goals: 12 Responsible Consumption and Production

Web of Science Categories

Engineering, Multidisciplinary; Materials Science, Composites

Funding

View funding text

Funding agency	Grant number	Show All Details
Horizon 2020	734272	Show details
UK Research & Innovation (UKRI) Engineering & Physical Sciences Research Council (EPSRC)	EP/ P025447/1	Show details
	EP/ P026087/1	Show details
UK Research & Innovation (UKRI)		Show details
Institution of Engineering and		

+ See more data fields

Journal information

COMPOSITES PART B-ENGINEERING

View Journal Impact

ISSN	1359-8368
eISSN	1879-1069
Current Publisher	ELSEVIER SCI LTD, 125 London Wall, London EC2Y 5AS, ENGLAND
Table of Contents	Current Contents Connect
Journal Impact Factor	Journal Citation Reports ™

14.2
Journal Impact Factor ™ (2024)

3.04
Journal Citation Indicator ™ (2024)

Research Areas	Engineering; Materials Science
Web of Science Categories	Engineering, Multidisciplinary; Materials Science, Composites

Citation Network

In Web of Science Core Collection

22
Citations

Create citation alert

22
Times Cited in All Databases

+ See more times cited

83
Cited References

→ View Related Records

How does this document's citation performance compare to peers?

← Open comparison metrics panel

Use in Web of Science

20	23
Last 180 Days	Since 2013

Learn more →

This record is from:

Web of Science Core Collection

- Science Citation Index Expanded (SCI-EXPANDED)

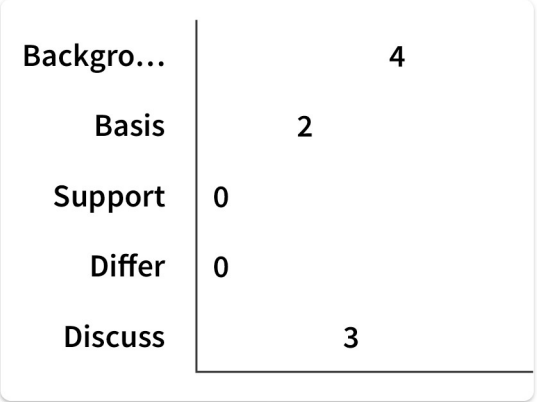
Suggest a correction

If you would like to improve the quality of the data in this record, please Suggest a correction

Data is from InCites Benchmarking & Analytics

Citing items by classification

Breakdown of how this article has been mentioned, based on available citation context data and snippets from 9 citing item(s).



Most Recently Cited by

Wang, XW; Wang, ZH; Shahzad, MM; et al.
An incremental broad ensemble learning framework for the identification of structural damage in conditions of data scarcity and non-stationarity
ENGINEERING STRUCTURES

Shen, JH; Xiang, S; Qin, Y; et al.
Progressive deep feature learning network based on fault-aware deformable convolution and its application in railway defect visual inspection
ENGINEERING APPLICATIONS OF

ENGINEERING APPLICATIONS OF ARTIFICIAL INTELLIGENCE

See all →

83 Cited References

View as set of results

Explore

Visualization includes 2 reference(s) not mentioned in the body of the article.

Showing 83 of 83 First appearance *(from Web of Science Core Collection)*

1 **Damage mechanism of carbon fiber reinforced plastic pipe based on reverse and forward curvature drilling**

Li, SJ; Zou, S; (...); Ko, TJ
Jul 15 2022 | COMPOSITE STRUCTURES 292

23 Citations

52 References

Enriched Cited References